tf.compat.v1.distributions.Categorical Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/distributions/Categorical

> *Categorical distribution.*

Function Signatures

```
tf.compat.v1.distributions.Categorical( logits=None,
probs=None, dtype=tf.dtypes.int32, validate_args=False,
allow_nan_stats=True, name='Categorical' )
```

Code Examples

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logits=None,
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```
tf.dtypes.int32
```

```
pmf(k; pi) = prod_j pi_j**[k == j]
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```
K <= min(2**31-1, {
tf.float16: 2**11,
tf.float32: 2**24,
tf.float64: 2**53 }[param.dtype])
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```

```
dist = Categorical(probs=[0.1, 0.5, 0.4])
n = 1e4
empirical_prob = tf.cast(
tf.histogram_fixed_width(
dist.sample(int(n)),
[0., 2],
nbins=3),
dtype=tf.float32) / n
# ==> array([ 0.1005,  0.5037,  0.3958], dtype=float32)
```

Documentation

Categorical distribution.

Inherits From: Distribution

The Categorical distribution is parameterized by either probabilities or log-probabilities of a set of K classes. It is defined over the integers {0, 1, ..., K}.

The Categorical distribution is closely related to the OneHotCategorical and Multinomial distributions. The Categorical distribution can be intuited as generating samples according to argmax{ OneHotCategorical(probs) } itself being identical to argmax{ Multinomial(probs, total_count=1) }.

Mathematical Details

The probability mass function (pmf) is,

Pitfalls

The number of classes, K, must not exceed:

- the largest integer representable by self.dtype, i.e.,

2**(mantissa_bits+1) (IEEE 754),

- the maximum Tensor index, i.e., 2**31-1.

In other words,

Examples

Creates a 3-class distribution with the 2nd class being most likely.

Creates a 3-class distribution with the 2nd class being most likely. Parameterized by logits rather than
probabilities.

Creates a 3-class distribution with the 3rd class being most likely.
The distribution functions can be evaluated on counts.

Attributes

Stats return +/- infinity when it makes sense. E.g., the variance of a Cauchy distribution is infinity. However, sometimes the statistic is undefined, e.g., if a distribution's pdf does not achieve a maximum within the support of the distribution, the mode is undefined. If the mean is undefined, then by definition the variance is undefined. E.g. the mean for Student's T for df = 1 is undefined (no clear way to say it is either + or - infinity), so the variance = E[(X - mean)**2] is also undefined.

May be partially defined or unknown.

The batch dimensions are indexes into independent, non-identical parameterizations of this distribution.

May be partially defined or unknown.

Currently this is one of the static instances distributions.FULLY_REPARAMETERIZED or distributions.NOT_REPARAMETERIZED.

Methods

batch_shape_tensor

View source

Shape of a single sample from a single event index as a 1-D Tensor.

The batch dimensions are indexes into independent, non-identical parameterizations of this distribution.

View source

Cumulative distribution function.

Given random variable X, the cumulative distribution function cdf is:

View source

Creates a deep copy of the distribution.

covariance

View source

Covariance.

Covariance is (possibly) defined only for non-scalar-event distributions.

For example, for a length-k, vector-valued distribution, it is calculated as,

where Cov is a (batch of) k x k matrix, 0 <= (i, j) < k, and E denotes expectation.

Alternatively, for non-vector, multivariate distributions (e.g., matrix-valued, Wishart), Covariance shall return a (batch of) matrices under some vectorization of the events, i.e.,

where Cov is a (batch of) k' x k' matrices,
0 <= (i, j) < k' = reduce_prod(event_shape), and Vec is some function mapping indices of this distribution's event dimensions to indices of a length-k' vector.

cross_entropy

View source

Computes the (Shannon) cross entropy.

Denote this distribution (self) by P and the other distribution by Q. Assuming P, Q are absolutely continuous with respect to one another and permit densities p(x) dr(x) and q(x) dr(x), (Shanon) cross entropy is defined as:

where F denotes the support of the random variable X ~ P.

entropy

View source

Shannon entropy in nats.

event_shape_tensor

View source

Shape of a single sample from a single batch as a 1-D int32 Tensor.

is_scalar_batch

View source

Indicates that batch_shape == [].

is_scalar_event

View source

Indicates that event_shape == [].

kl_divergence

View source

Computes the Kullback--Leibler divergence.

Denote this distribution (self) by p and the other distribution by q. Assuming p, q are absolutely continuous with respect to reference measure r, the KL divergence is defined as:

where F denotes the support of the random variable X ~ p, H[., .] denotes (Shanon) cross entropy, and H[.] denotes (Shanon) entropy.

log_cdf

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Log cumulative distribution function.

Given random variable X, the cumulative distribution function cdf is:

Often, a numerical approximation can be used for log_cdf(x) that yields a more accurate answer than simply taking the logarithm of the cdf when x << -1.

log_prob

View source

Log probability density/mass function.

log_survival_function

View source

Log survival function.

Given random variable X, the survival function is defined:

Typically, different numerical approximations can be used for the log survival function, which are more accurate than 1 - cdf(x) when x >> 1.

View source

View source

param_shapes

View source

Shapes of parameters given the desired shape of a call to sample().

This is a class method that describes what key/value arguments are required to instantiate the given Distribution so that a particular shape is returned for that instance's call to sample().

Subclasses should override class method _param_shapes.

param_static_shapes

View source

param_shapes with static (i.e. TensorShape) shapes.

This is a class method that describes what key/value arguments are required to instantiate the given Distribution so that a particular shape is returned for that instance's call to sample(). Assumes that the sample's shape is known statically.

Subclasses should override class method _param_shapes to return constant-valued tensors when constant values are fed.

View source

Probability density/mass function.

quantile

View source

Quantile function. Aka "inverse cdf" or "percent point function".

Given random variable X and p in [0, 1], the quantile is:

sample

View source

Generate samples of the specified shape.

Note that a call to sample() without arguments will generate a single sample.

stddev

View source

Standard deviation.

Standard deviation is defined as,

where X is the random variable associated with this distribution, E denotes expectation,